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Smart Filter Performance Monitoring System

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ABSTRACT

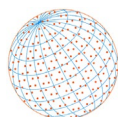
Air filters are widely used in residential and industrial applications. It is designed to remove particulate pollutants in the air to supply cleaner air to either occupants or industrial equipment. Without air filters, the occupants might suffer from polluted air, and expensive industrial equipment could be damaged by contaminants. However, air filters are installed and operated with a limited performance monitoring system, and the efficacy of air filters is unknown after replacement. Here, we propose and prototype a smart filter performance monitoring system that costs less than 200 U.S. dollars and can report filtration efficiency, differential pressure, temperature, and RH (relative humidity) in real time. Three case studies are presented: an air purifier, a teaching building HVAC (heating, ventilation, and air conditioning) system, and a large-scale air cleaning system. Applications of the proposed system cover from household to industrial, which not only verified the concept of the system but also proved its feasibility and advantages of the system. With the proposed system, residential customers can rest assured with their air purifiers or HVAC furnace filter; industrial customers can monitor the filtered air cleanness that will enter their internal combustion engines or gas turbines. Moreover, filter monitoring data can establish a database for researchers to validate the filter models or train a machine learning model for filter performance prediction. Filter or air-cleaning device manufacturers can improve their products or recommend suitable products to their customers based on big data contributed by this filter monitoring system.

Keywords: Filter performance, Filtration efficiency, Low-cost sensors, Air quality monitoring, Air filter

1 INTRODUCTION

Air filters are currently installed and operated with limited performance monitoring systems, such as filter differential pressure. Meanwhile, the criteria for filter replacements are based on the installed time or mileage, for example, three months for residential HVAC (heating, ventilation, and air conditioning) filters and ~12000 miles for vehicle cabin air filters. However, the actual filter operating time and pollutant concentration are not considered. Air filters are commonly used for a prolonged time or cleaned with compressed air and then reused. However, the performance of air filters is not closely monitored, especially the efficiency.

HVAC Air filters are designed to protect people staying indoors, and occupants' health is invaluable. It has been a rising concern that the HVAC system may transmit COVID-19, and ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) recommends that MERV-13 or above level filtration should be used when it is achievable ([ASHRAE, 2020](#)). The purpose of using high-level filters is to capture virus aerosol in room air, whereas some HVAC engineers worry the improper installation of air filters or the leaks around filter edges may jeopardize the overall filtration efficacy. Improper installation or defective filters can give people a false sense of safety, which is not desired. Therefore, ensuring the HVAC filtration system is operating as designed is critical. Gas turbines, internal combustion engines, and dust collection systems all use air filters. Compared to the equipment behind air filters, the price of air filters is negligible, not to mention the repair cost if the equipment is damaged due to defective air filters or improper installations. It is unclear whether air filters are functioning as designed without



sufficient air filter performance information. When switching to a new filter brand, it is unrealistic for end-users to judge the quality of the new filters. The uncertainties of air filter performance make people or equipment behind the air filters more vulnerable. On the other hand, the lack of information could cause early or prolonged filter replacements. When air filters are not replaced confidently, it could be either a waste of filters or loss of protection, or a waste of energy to overcome extra filter differential pressure.

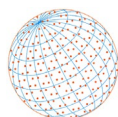
As concluded from the examples above, filter performance monitoring systems are highly demanded. Currently, there are several commercially available filter monitoring sensors. Table 1 lists four sensors that can monitor air filters. In fact, all four sensors are based on differential pressure measurements, and none of them could measure filter efficiencies, temperature, and RH (relative humidity). 3M smart filter and Clean Alert filter monitor are designed for residential HVAC applications. 3M smart filter uses Bluetooth to connect with the customer’s cell phones, alerting when it is time to replace the filter. However, the smart sensor is glued on the HVAC filter and cannot be reused. Therefore, its cost is higher than conventional HVAC filters. The Clean Alert filter monitor can only send a message to customers when it is time to replace the HVAC filters. The other two industrial filter monitoring sensors have online monitoring functions with a paid subscription. The Donaldson iCue is designed for dust collectors, which gives the differential pressure measurement to monitor and predict the filter’s life. Mann+Hummel Senzit is designed for the truck intake air filter monitoring installed downstream of the intake air filter. The data is transmitted through the cellular network, and the truck’s location is also reported. It is designed for fleet managers to arrange maintenance in the fleet base to reduce the fleet operation cost.

The commercially available sensors are not capable of RH and temperature measurements. In addition, filter efficiency is also not reported. The actual protection that filters can offer is unclear. With the current monitoring methods, it is doubtful to detect improper filter installation and defective filters. Therefore, designing a filter monitoring system with filter efficiency measurement capability is vital. With that, filter operation will not be a “black box”, and the protections that filters can offer can be reported. At the same time, the comprehensive filter operation condition could be recorded, which could be used to predict the filter life more accurately. For example, the temperature and RH effect on the filter life will be discussed in Part 2 later.

Numerous studies have developed filter testing/monitoring systems using research-grade particle instruments (Cao *et al.*, 2023; Chen *et al.*, 2022; Lyu *et al.*, 2021; Ou *et al.*, 2020, 2017; Chen *et al.*, 2016; Tian *et al.*, 2021; Pei *et al.*, 2019a). By measuring the particle concentrations upstream and downstream of the filter media, filter filtration efficiencies can be obtained (Pei *et al.*, 2020; Pan *et al.*, 2023; Pei *et al.*, 2021, 2019b). However, those monitoring systems are bulky, expensive, and designated for research purposes. On the other hand, the increasingly available low-cost light-scattering particulate matter (PM) sensors significantly impact the air market as they offer a cost-effective solution for air quality monitoring. A low-cost light-scattering PM sensor typically consists of a light source (laser diode, LED) to send a light beam to illuminate a small volume of aerosol particle stream that passes the viewing zone. A part of the light is scattered and collected by a detector. The simplicity of the design leads to low cost. The current low-cost PM sensor studies focus on indoor (Chen *et al.*, 2021; Levy Zamora *et al.*, 2019; Olivares *et al.*, 2012), outdoor (Hong *et al.*, 2021; Johnson *et al.*, 2018; Malings *et al.*, 2020), or spatiotemporal PM measurements (Feinberg *et al.*, 2019; Li *et al.*, 2018; Patel *et al.*, 2017), however, their potential for filter filtration performance monitoring has not yet been demonstrated. Given the lack of commercially available

Table 1. Comparison of current commercially available products.

Product	Application	Pressure Transducer	Online Monitoring	RH, T Monitoring	Filter Efficiency Monitoring
3M smart filter	Residential HVAC	Single	No Bluetooth only	No	No
Clean Alert filter monitor	Residential HVAC	Single or Differential	No	No	No
Donaldson iCue	Industrial dust collector	Differential	Yes Paid Subscription	No	No
Mann+Hummel Senzit	Industrial Fleet	Single	Yes Paid Subscription	No	No



products or studies on filter performance monitoring, this study developed a portable smart filter monitor system using off-the-shelf, low-cost light scatter particle sensors and RH, temperature, and pressure sensors. This study aims to demonstrate the capability of low-cost sensors in filter performance monitoring applications and the importance of monitoring filter performance.

2 SYSTEM DESCRIPTION

To measure the filter efficiency, the particle concentrations at the upstream and downstream of the filter are needed, as shown below:

$$\text{Efficiency} = 1 - \left(\frac{\text{Downstream concentration}}{\text{Upstream concentration}} \right) \quad (1)$$

Similarly, the differential pressure can be measured by a pair of pressure sensors. Fig. 1 shows the schematic of the filter performance monitoring system. Upstream and downstream measurement signals are processed by an Arduino microcontroller, and an I²C multiplexer is employed to switch the I²C addresses of each sensor. Currently, the filter operating parameters are uploaded to thingspeak.com via Wi-Fi, an open platform for IoT devices. Thingspeak.com can be accessed from a computer, a laptop, a tablet, or a cellphone. Therefore, the filter operating parameters are available wherever there is internet access. With the use of Cloud storage, this filter monitoring system allows for establishing a filter operating parameter database that includes filtration efficiency, which is valuable for researchers to do filter modeling and for filter manufacturers to improve their filter designs. It's also proposed to use the filter monitoring data to train a machine learning model integrated into the web interface to predict user filter performance in certain circumstances. This will be done in the later custom-made web interface version as a part of the filter performance monitoring system.

The microcontroller used in this system is Arduino Uno Wi-Fi R2, the most common microcontroller for electronic prototyping. The Arduino Uno Wi-Fi R2 has a built-in Wi-Fi module to connect to the internet. The communications between the microcontroller and sensors are accomplished by I²C protocol. Each of the sensors has a fixed I²C address. Since particle and environmental sensors are used in pairs, the I²C addresses will be duplicated. An I²C multiplexer is employed to switch the communications between sensors so that the data transmission is achieved sequentially.

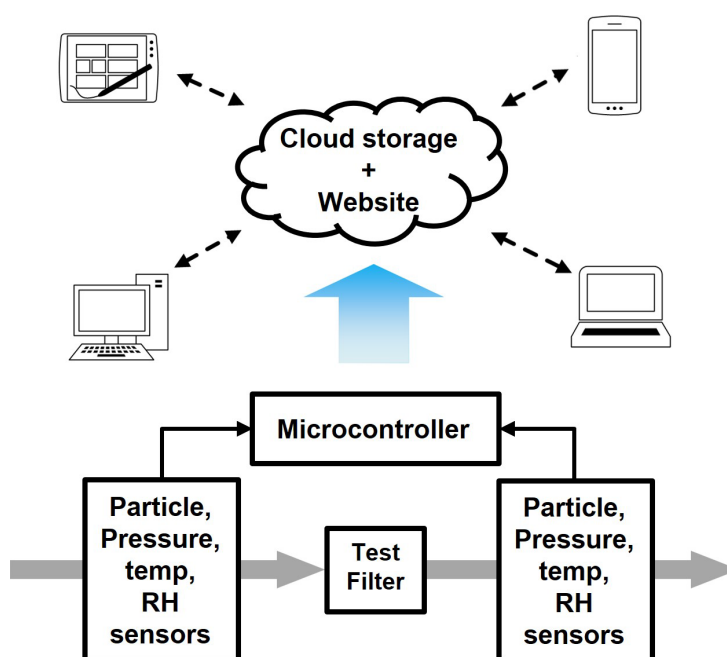
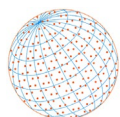


Fig. 1. Schematic of the filter performance monitoring system.



The particle sensor used in this system is Sensirion SPS30, a commercially available optical particulate matter sensor. According to the manufacturer, the innovative contaminant-resistant technology can keep the detection chamber clean to report accurate concentration throughout its life for at least eight years of continuous operation. The mass concentration limit is $1000 \mu\text{g m}^{-3}$, which is high enough for the ambient pollutant application. Another advantage of the SPS30 is that it can offer 4-channel particle size concentrations, $\text{PM}_{1.0}$, $\text{PM}_{2.5}$, $\text{PM}_{4.0}$, and PM_{10} . The particle size information is critical to characterize the particulate pollutants and monitor the filter performance.

The environmental sensor used in this system is Bosch BME680, a multifunction sensor. It can measure temperature, RH, pressure, and breathable VOC. The RH detection range is 0–100%; The temperature detection range is -25°C – 85°C ; The pressure range is 300 hPa–1100 hPa. All three parameter’s operating ranges are suitable for ambient air measurement. At the downstream of the filter, the pressure is lower than the ambient pressure, so this sensor is suitable for absolute pressure measurements. This sensor has a small footprint of $3.0 \text{ mm} \times 3.0 \text{ mm}$, and it is a surface-mount device that can be mounted on a customized printed circuit board in a later design. In the current design, a sensor module with a larger footprint is used for prototyping purposes.

A photo of the filter performance monitoring system is shown in Fig. 2. An additional display is installed to show the operating status. The one shown in the photo is made to compare the sensor readings parallelly. All readings are within 2.5% error; hence, sensors are interchangeable as claimed.

The cost of the filter monitoring system is another aspect worth considering. Compared to the filter efficiency test setup in the laboratory or the testing facility that costs tens of thousands of dollars, the total cost of this monitoring system is only about 190 USD (U.S. dollars). The bill of material of the current prototype is listed in Table 2. Unlike some sensors attached to air filters, this filter performance monitoring system doesn’t need extra replacement except battery. The total cost can be regarded as the initial investment, which can be averaged for the life of the system. It should be noticed that the Arduino is used here since it is a common platform for

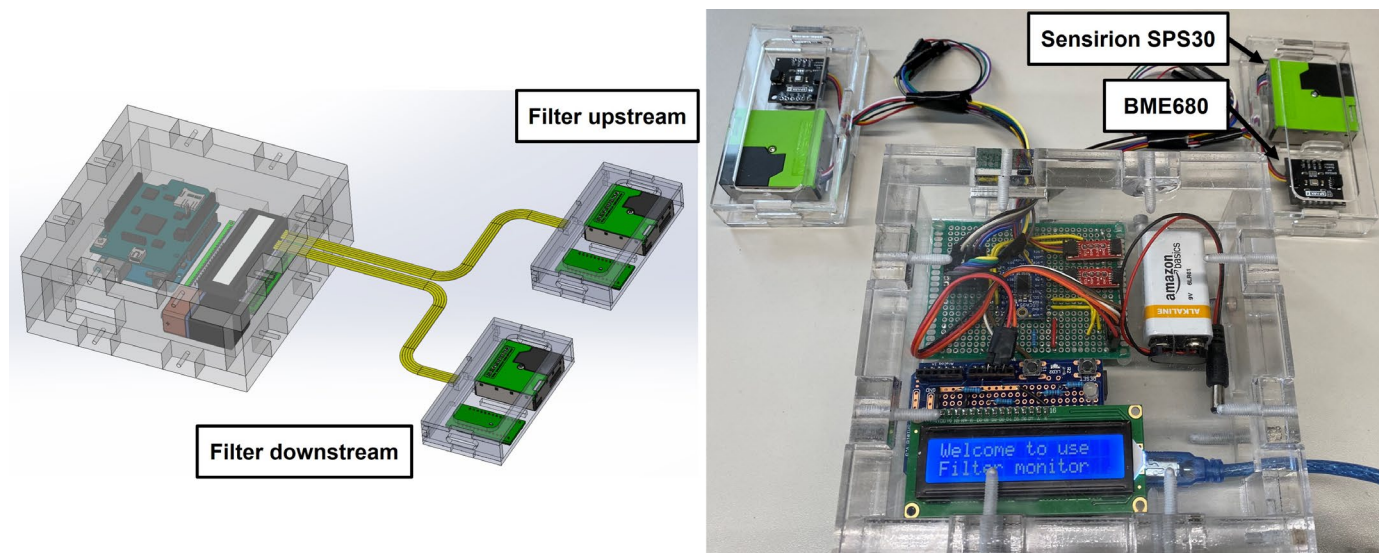
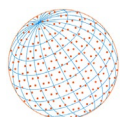


Fig. 2. Photo of the filter performance monitoring prototype.

Table 2. Bill of material of the current prototype.

Item	Price/unit (USD)	Quantity	Notes
Arduino Uno Wi-Fi R2	44.95	1	Microcontroller
Sensirion SPS30	46.95	2	Particle sensor
Bosch BME680	19.95	2	Environmental sensor
TCA9548A I ² C Multiplexer	6.95	1	Communication component
Other (wires, Perfboard)	5		
Total Cost	190.7		



prototyping conveniently, and the BME680 used here is also mounted on a prototype circuit board. The actual cost of the filter performance monitoring could be reduced if Arduino is replaced with ESP32 and a customized printed circuit board is designed for ESP32 and BME680. In that way, the total cost could be below 120 USD. The design of a second-generation filter monitoring system that is currently under development can be found in the [Supplemental Material](#).

Next, three case studies will be covered here to project potential applications of this smart filter performance monitoring system.

3 APPLICATION EXAMPLES AND RESULTS

3.1 Case Study 1: Indoor Air Purifier

Indoor air purifiers are common residential air cleaning devices. This prototype is installed on an indoor air purifier (Oreck Air Instinct 200, Cookeville, TN) to monitor the HEPA filter performance. [Fig. 3](#) shows how the filter performance monitoring system is installed on the air purifier. The Arduino and other circuit boards are enclosed in an acrylic box, which is mounted on the front panel of the indoor air purifier. The upstream module is mounted inside the front panel to measure the operating parameters of the HEPA filter, which measures the indoor air quality, absolute pressure, temperature, and RH. Behind the HEPA filter, the downstream module is installed to detect the air quality, temperature, and RH after HEPA filtration. In [Fig. 3](#), the HEPA filter is not in place so that the downstream module can be seen. It was found that the HEPA filter paper frame does not seal tightly, and thus extra foam weather strip is applied around the HEPA paper frame to improve the sealing between the HEPA filter and the air purifier.

The sample readings of this system are listed in [Table 3](#), where the air purifier is set at medium fan speed. Both mass and number concentrations of $PM_{1.0}$, $PM_{2.5}$, $PM_{4.0}$, and PM_{10} are reported. It can be found that for the sample readings shown below, the overwhelming particles are $PM_{1.0}$, with limited $PM_{2.5}$ particles, as the readings of $PM_{2.5}$, $PM_{4.0}$, and PM_{10} are the same. Since PM_{10} concentrations represent all detected particles, both PM_{10} mass and number efficiencies are calculated in [Table 3](#). The mass and number efficiencies are slightly less than the nominal HEPA efficiency of 99.97%, which might be due to leaks around the HEPA filter, though the weather strip applied at the sides of the filter. This is an excellent example: even with careful sealing, the overall filtration efficacy is less than the designed filtration efficiency. The non-sealed filtration efficiency will be discussed later in this section. In addition, the SPS30 sensor can also report the

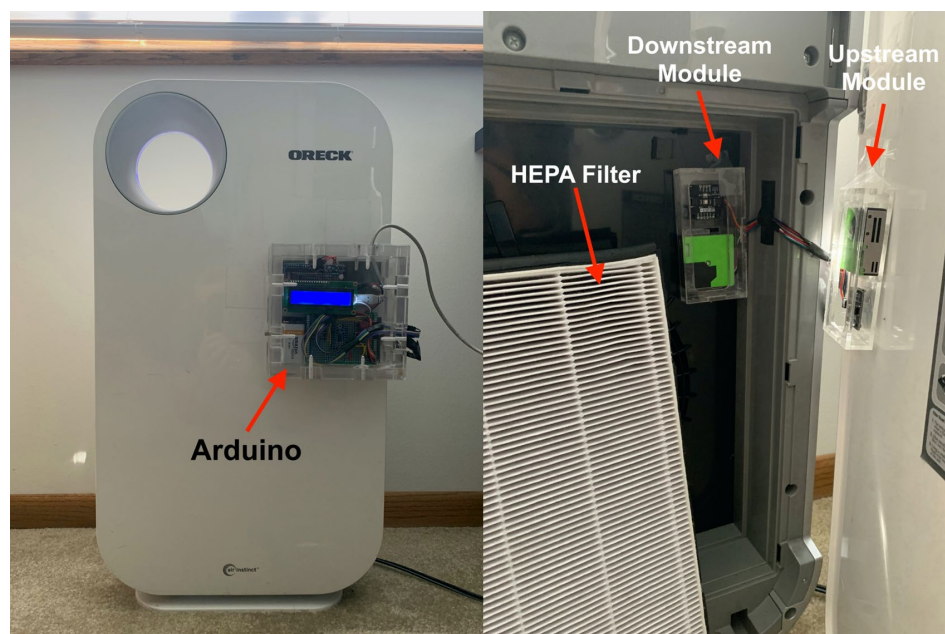


Fig. 3. The filter performance monitoring system on an indoor air purifier (HEPA filter is not in place to show the downstream module).

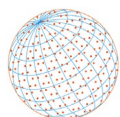


Table 3. Sample readings of the filter performance monitoring system at medium fan speed.

Upstream					
Number Concentration (# cm ⁻³)					
PM _{1.0}	PM _{2.5}	PM _{4.0}	PM ₁₀	Average Particle Size	0.47 μm
14.51	14.61	14.61	14.61	Pressure	97878 Pa
Mass Concentration (μg m ⁻³)				Temperature	25.63°C
PM _{1.0}	PM _{2.5}	PM _{4.0}	PM ₁₀	Relative Humidity	64.09%
1.83	1.94	1.94	1.94		
Downstream					
Number Concentration (# cm ⁻³)					
PM _{1.0}	PM _{2.5}	PM _{4.0}	PM ₁₀	Average Particle Size	0.39 μm
0.28	0.28	0.28	0.28	Pressure	97834 Pa
Mass Concentration (μg m ⁻³)				Temperature	25.9°C
PM _{1.0}	PM _{2.5}	PM _{4.0}	PM ₁₀	Relative Humidity	63.2%
0.03	0.04	0.04	0.04		
PM ₁₀ Number Efficiency		PM ₁₀ Mass Efficiency		Differential Pressure	
98.08%		97.94%		44 Pa	

average particle size, which is 0.47 μm and 0.39 μm upstream and downstream, respectively. Thus, a slight particle size distribution shift occurred during filtration. The indoor air PM_{2.5} mass concentration is 1.94 μg m⁻³, far below the EPA annual PM_{2.5} standard of 12 μg m⁻³. In contrast, the PM_{2.5} mass concentration downstream of the air purifier is only 0.04 μg m⁻³, indicating the effectiveness of an air purifier in removing indoor particulate matter. In the meantime, the absolute pressures, temperatures, and RHs are also reported. The HEPA filter differential pressure could be calculated based on the pressure data, which is 44 Pa.

The air purifier used here has four fan speeds, low (100 CFM), medium (180 CFM), high (240 CFM), and turbo (350 CFM). With the same filtration area, the higher the fan speed, the higher the differential pressure. A comparison of differential pressures at different fan speeds is shown in Fig. 4. The average differential pressures for each fan speed are calculated from 75 sample points when the fan speed is stable. It was found a noticeably clear differential pressure raised with increasing fan speed. As mentioned before, the HEPA filter does not fit the air purifier tightly. Non-sealed differential pressures at various fan speeds are also shown in Fig. 4. It is reasonable that differential pressures of non-sealed are lower than the sealed conditions due to leaking points. However, differential pressure can only verify the installation when the leak is significant enough. It is impossible to tell whether the HEPA filter is sealed and whether the HEPA filter is performing as expected only by comparing the differential pressure information.

To verify that the HEPA filter is performing as expected, the filtration efficiency is the only criterion. Fig. 5 compares the mass and number efficiencies of PM₁₀ at various fan speeds. Significant

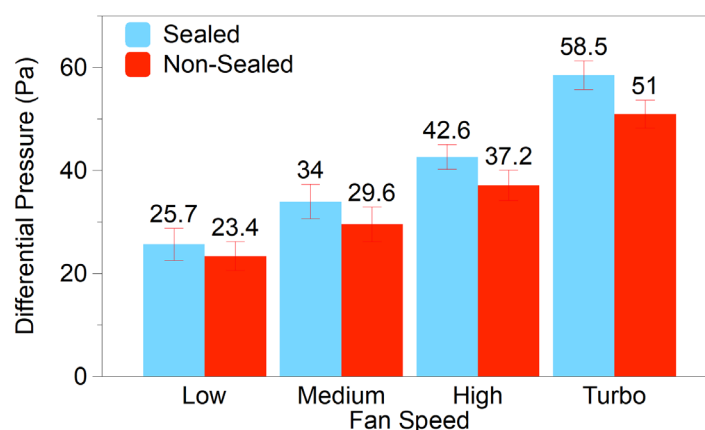


Fig. 4. Differential pressures of the HEPA filter at different fan speeds.

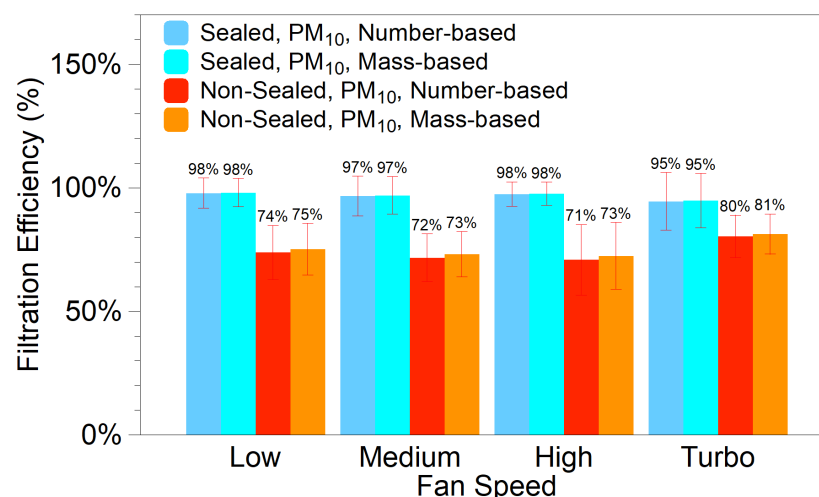
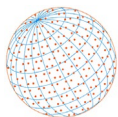


Fig. 5. Filtration efficiencies of the HEPA filter at different fan speeds.

efficiency discrepancies show that the non-sealed HEPA filter in the air purifier can only achieve about 70%–80% overall filtration efficacy. The clean air delivery rate is overestimated when the actual filtration efficacy is below the designed. This condition may also happen to the residential HVAC furnace filters, whose frames are also made of cardboard without sealing foam around them. Hence the protection offered by the HVAC furnace filters was also overestimated due to the imperfect installation. Since indoor air purifiers and residential HVAC furnace filters circulate indoor air in a room that can be considered a closed space, the deficient clean air delivery capability can be mitigated by extra circulation and longer operating time. However, when air filters handle air with the “single pass” application, such as the engine intake air filter, the deficient filtration capability can result in severe damage to the engine.

From the case study of the indoor air purifier, it can be found that the differential pressure measurement cannot offer sufficient information to verify that filters are performing as designed and there are potential hazards to occupants or equipment behind air filters. A pair of particle sensors across the air filter can support a confident operation with filtration efficiency reported. Or at least a single particle sensor downstream of air filters can offer clean air delivery capability when absolute cleanness is required.

3.2 Case Study 2: Teaching Building HVAC Monitoring System during COVID-19

HVAC (Heating, Ventilation, Air Conditioning) system is the system that provides thermal comfort for everyone inside residential or commercial buildings and exchanges indoor air with fresh outdoor air. During the air exchange process, so-called ventilation, the MERV (Minimum Efficiency Reporting Values) rating filters are used to remove airborne particulate contaminants, which is critical for improving indoor air quality. During the COVID-19 pandemic, the filtration performance of the HVAC system caught more attention since the particulate contaminants can carry the virus. In this case study, we placed the filter monitor prototypes upstream and downstream of the pre-filter (MERV 8) and final filter (MERV 15) in AHU (air handling unit, used for air ventilation) in a university teaching building to measure whether the filter performs as of expectation. MERV rates the filter filtration performance between 0.3 and 10 microns. The detailed efficiency parameters are listed in Table 4.

Fig. 6 shows the sensor placement in the HVAC system for filtration performance monitoring. The filter monitors were placed at four locations: upstream of the pre-filter MERV 8, downstream of the pre-filter MERV 8 (namely, upstream of the final filter MERV 15), downstream of the final filter MERV 15 in the rooftop AHU, and the AHU ventilation inlet in a modern classroom in the teaching building. Note that it was difficult to run the sensor wires upstream and downstream of the filter in this large space application. Thus, instead of using the one controller with two pairs of sensors prototype configuration described above, the one controller with one pair of sensors configuration prototype described in detail in the previous study (Chen *et al.*, 2021) was used.

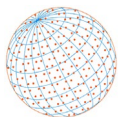


Table 4. Minimum Efficiency Reporting Values (MERV) Parameters (ASHRAE, 2017).

ASHRAE 52.2 MERV	Composite Average Particle Size Efficiency, % in size range, μm		
	Range 1 0.3–1.0	Range 2 1.0–3.0	Range 3 3.0–10.0
8	N/A	$20 \leq E_2$	$70 \leq E_3$
15	$85 \leq E_1$	$90 \leq E_2$	$95 \leq E_3$

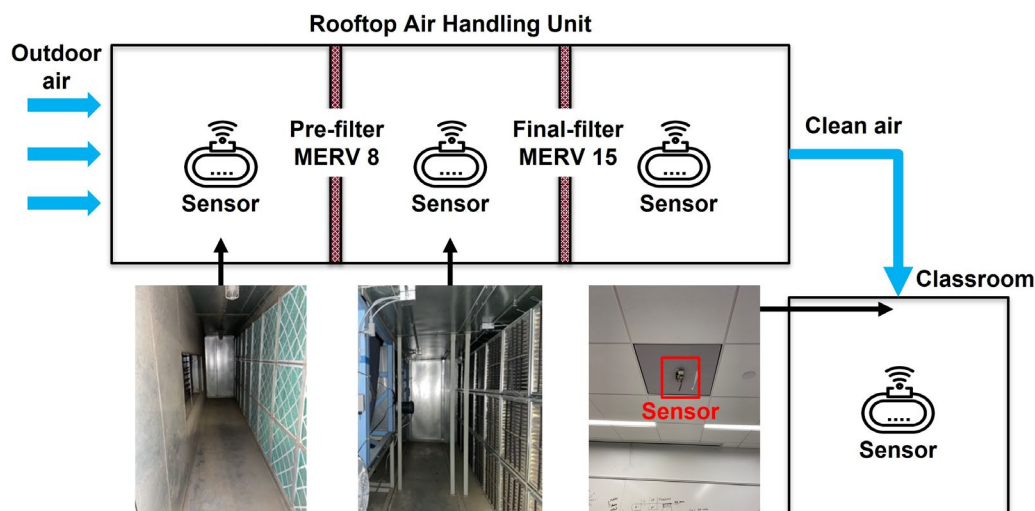


Fig. 6. Sensor placement for the HVAC filter performance monitoring.

So, one prototype unit was placed at one location of interest to eliminate the running wire in between. The particle concentrations at different locations will be compared to obtain the filter performance, thus the same concept. The AHU was operated under 300 CFM airflow volume.

Fig. 7 reports the one-minute average measured $\text{PM}_{2.5}$ particle concentrations at the four locations. Two repeats were done on different days at different outdoor concentration levels. In the first repeat, the outdoor concentration was relatively low and stable. The concentration decreased as the airflow progressed through filter chambers in the AHU unit. The concentration at the AHU ventilation inlet in the classroom was measured slightly lower than the concentration downstream of the AHU final filter. Since there was no other filtration system between these two locations, their concentrations should have a fair agreement. The slightly lower concentration in the classroom was possibly due to the particle loss during the transportation, which was of expectation. These results also validated the reliability of the particle measurements. By comparing the measured upstream and downstream mass concentrations, the mass-based filtration efficiencies of pre-filter MERV 8 and final-filter MERV 15 were obtained (Eq. (1)) with mean values of 25% and 36%, respectively. The results are consistent with the MERV rating that the MERV 15 filter should perform better than the MERV 8. However, their efficiencies were much lower than expected. Note that the PM sensors used here were tested and showed good consistency between sensors and good sensor response linearity at low concentrations in the previous study (Chen *et al.*, 2021). Also, since the efficiency measurement uses the concentration ratio, the uncertainty from the absolute concentration measurement was minimized. One may concern about the low measured efficiency because conventional light-scattering particle sensors have a small detection limit (Ye *et al.*, 2022), typically around 300 nm. The limit of detection was believed to have minimal influence on the results because the PM sensors measure mass concentration, and thus, the efficiency reported here is based on mass, and the small particles (< 300 nm) don't contribute too much mass. Moreover, the air purifier case study shows that filtration efficiency significantly improved and got close to its claimed values after improving the sealing. Therefore, it is reasonable to believe that the low filtration efficiencies found in the HVAC system were due to imperfect installation or possible system bypass leaking.

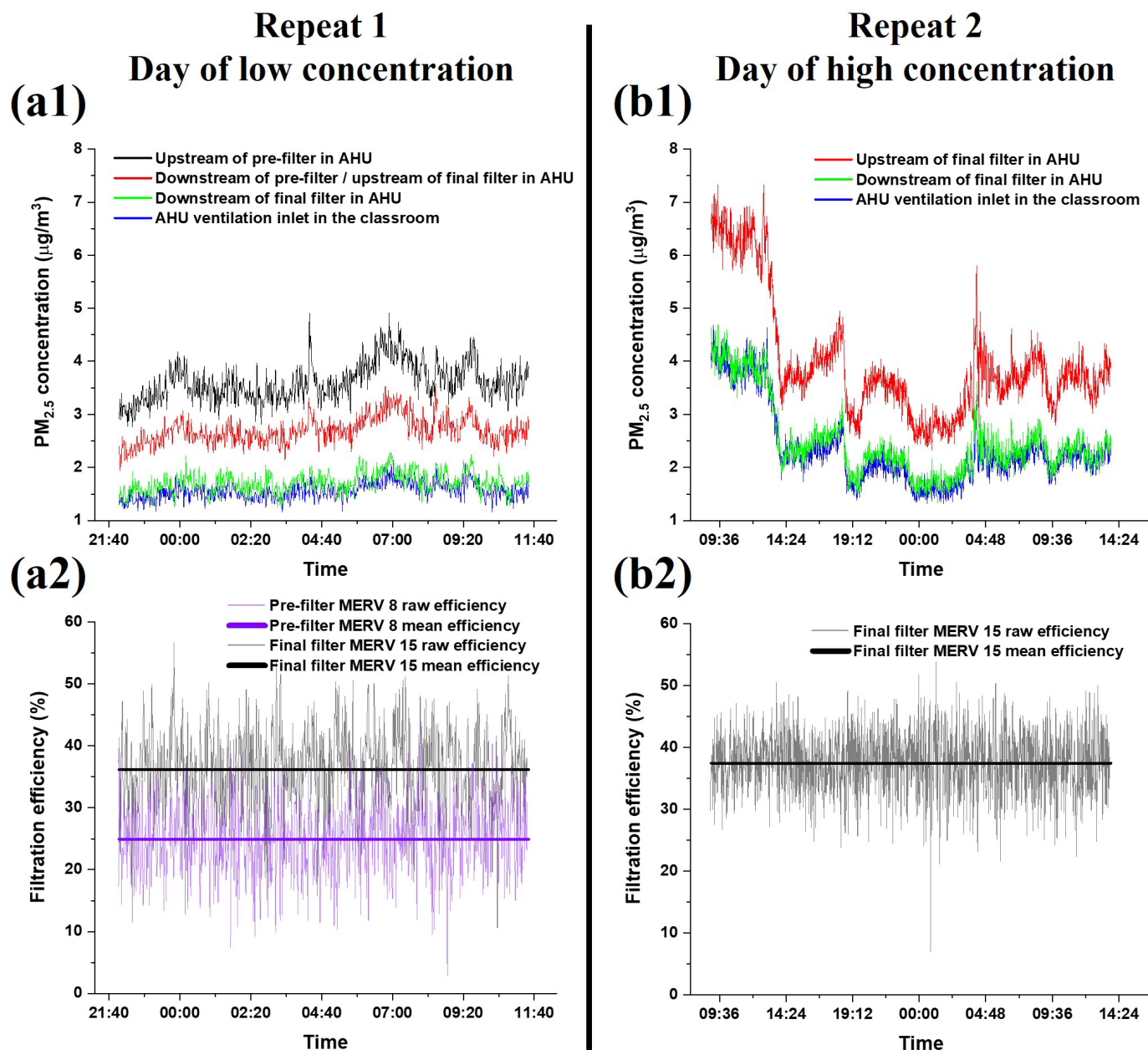
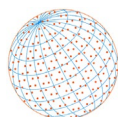
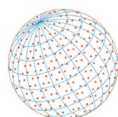


Fig. 7. Concentration and efficiency measurements using the HVAC monitoring system.

The second test repeat was done under the same AHU operation condition but on a day with higher outdoor concentration with a higher variation. The efficiency of the MERV 15 final filter was measured, averaging 37%, which agrees well with the first test results. Moreover, the concentration variation doesn't affect the efficiency, which suggests the PM sensors still gave a linear response in the tested concentration range, and the efficiency results were reliable. The concentration downstream of the AHU final filter was still slightly higher than the concentration in the ventilation inlet in the classroom, which is consistent with the first test results. There was no pre-filter upstream data in the second test, as there was a data transmission issue for the monitor in that location. The repeated test verified that the concentration effect and the repeatability of the efficiency measurement are within expectation at both low and high outdoor concentration levels.

The case study of the teaching building HVAC monitoring system shows that the filter monitoring system that utilizes a pair of particle sensors across the air filter can give reliable efficiency measurement in commercial building applications. The results show that the actual filtration performance could be much lower than the efficiency reported in the filter specifications due



to imperfect installation and/or HVAC system malfunctions, which suggests that it is necessary to install filter monitors in a building HVAC system to ensure the safety and health of the occupants.

3.3 Case Study 3: Delhi SALSCS Operation Monitoring

A solar-assisted large-scale cleaning system (SALSCS) was designed to handle regional air pollution (Cao *et al.*, 2018a, 2018b, 2015). The first generation of SALSCS works based on the stack effect, that the heated ground air under the transparent solar collector can flow to the chimney by passing the air filter bank. The ambient air can be cleaned to mitigate regional air pollution. In the second generation SALSCS, the airflow is reversed, and a fan is installed in the chimney to push the ambient air to the air filter bank and supply the clean air to the ground level. A demonstration system of SALSCS has been built in Xi'an, China.

The Indian air quality is a concern for public health. According to the data compiled by IQAir, among the 30 cities with the worst air pollution globally, 21 cities are from India (IQAir, 2019). Many urban areas in India are experiencing air pollution far higher than EPA standards (Guttikunda and Calori, 2013). To reduce the air pollution burden, the Indian government plans to reduce PM_{2.5} by 30% in 5 years. The SALSCS demonstration system in Xi'an proves that the SALSCS is a valuable tool for improving regional air quality. Hence Delhi is planning to build several second-generation SALSCS to comply with the 5-year air quality target. The filter performance monitoring system can be installed on the SALSCS to offer filter and system operating monitoring. The schematic diagram of the Delhi SALSCS and filter performance monitoring system is shown in Fig. 8.

The filter monitoring system is shown on the right side of Fig. 8, and Fig. 9 shows a detailed sensor location diagram. The sixteen sensor modules (particle sensor SPS30 + environmental sensor BME680, yellow circle) are installed upstream and downstream of the filter bank (green zigzag). Two pairs of sensor modules are on each side of the filter bank to monitor the filter performance (efficiency, differential pressure) because of the large area of the filter wall. Besides the filter performance monitoring system, five TSI DustTraks (8543, Shoreview, MN) are also installed around the SALSCS. There is one DustTrak inside the chimney to sample the ambient incoming air, and the other four are placed on each downstream side to sample the filtered air. The cost advantage of the described filter performance monitoring system is reflected in this case study. Sixteen sensor modules can be deployed, while only five DustTrak is planned due to the cost consideration. The more sample points, the more accurate the system operating information can be collected. In addition to the filter efficiency monitoring, the fan speed control is also integrated

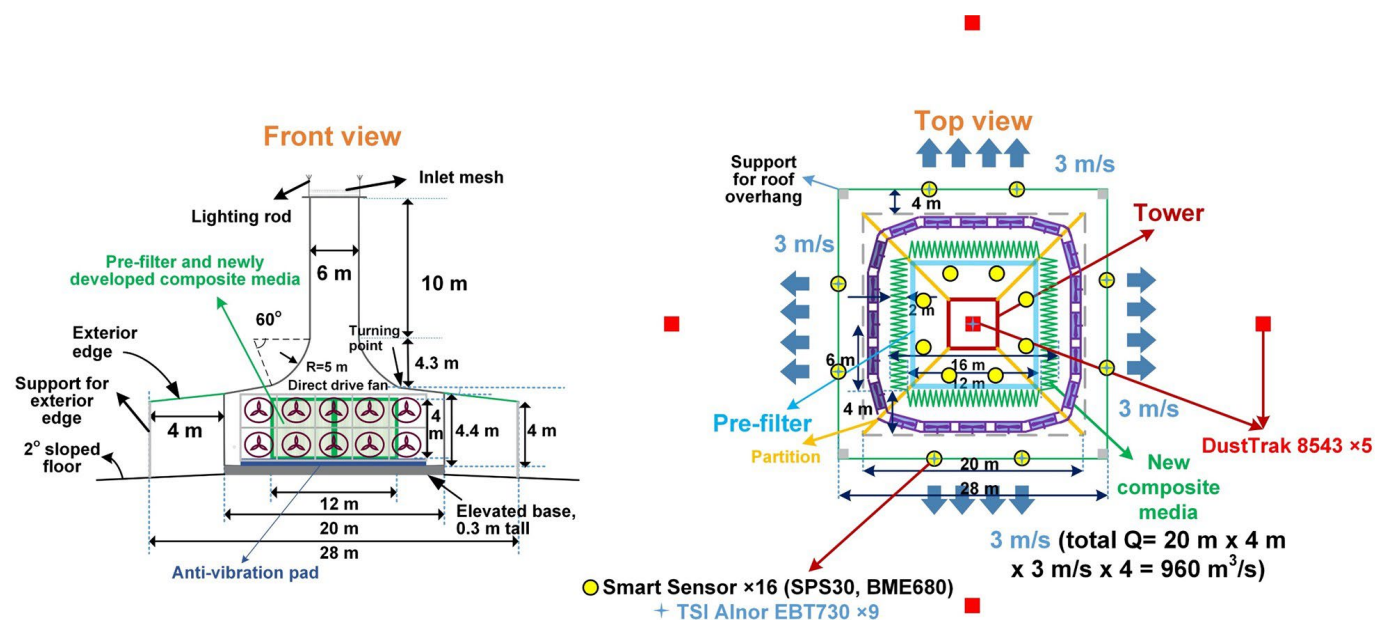
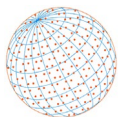


Fig. 8. Schematic diagram of the Delhi SALSCS and filter performance monitoring system (modified from Dr. Sheng-Chieh Chen's diagram).



Top view

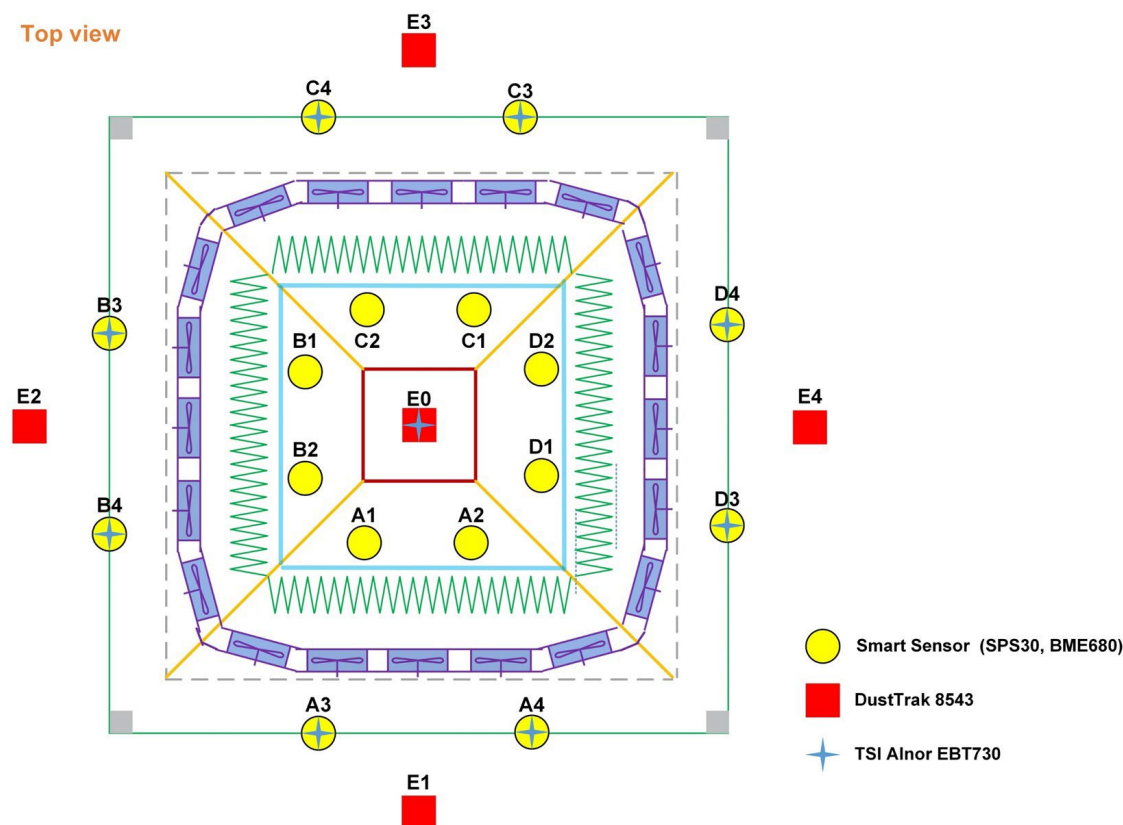


Fig. 9. Locations of filter performance monitoring system sensors and other measurement equipment in the Delhi SALSCS.

here. Nine TSI micromanometers (Alnor EBT730, Shoreview, MN) are proposed to measure the face velocity so that the fan speed can adjust accordingly.

The sensing and control system is shown in Fig. 10. With the sensors mentioned above, the filter differential pressure, filtration efficiency, and clean air delivery rate (CADR) can be calculated, while the fan speed can be adjusted based on the CADR requirement. With the filter performance monitoring system reporting the filter differential pressure and filtration efficiency, the SALSCS system can operate more efficiently and effectively, and the filter replacement can be done more confidently.

4 CONCLUSIONS

The present paper has described a filter performance monitor concept that uses low-cost sensors to offer the real-time filtration efficiency of air cleaning devices together with the differential pressure, temperature, and RH. Three case studies were conducted to project the potential applications and usefulness of this filter performance monitoring system. The case studies on indoor air purifiers and commercial building HVAC systems show that the filter performance monitoring system can enable reading of the filtration efficiency in addition to filter differential pressure which quantifies the protection an air cleaning device can offer. The results also demonstrated that the actual filtration performance of the air-cleaning device could be lower than expected due to improper installation. Hence, it's necessary to monitor the filter performance. The case study on Delhi SALSCS operation monitoring projects the application of the filter monitoring concept on the large-scale air cleaning device and the cost advantages of the proposed filter monitoring concept.

With the smart filter performance monitoring system, residential customers can rest assured with their air purifiers or HVAC furnace filter; industrial customers can monitor the filtered air cleanliness that will enter their internal combustion engines or gas turbines. Moreover, filter monitoring data can be used to establish a database for researchers to validate the filter models

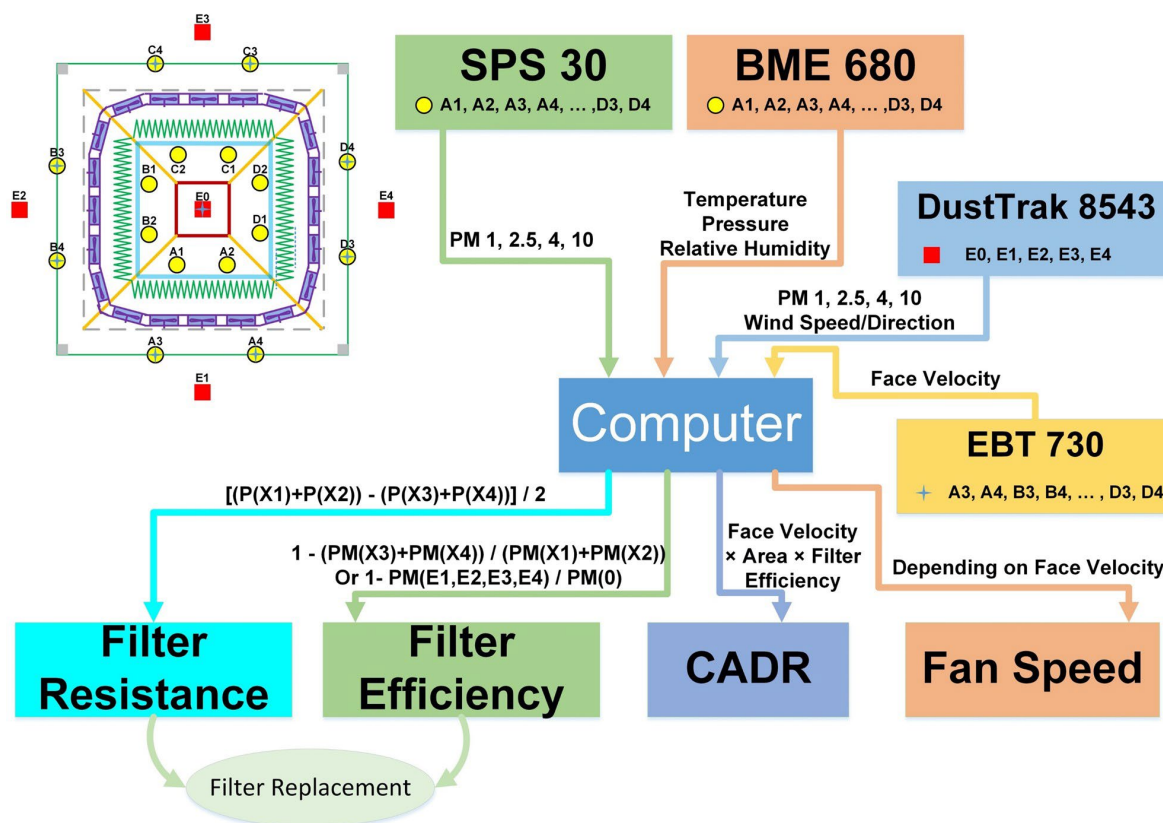
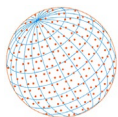


Fig. 10. Sensing and control system diagram of the Delhi SALSCS.

or train a machine learning model for filter performance prediction. Based on the big data from the smart filter monitoring system, the filters or air-cleaning device manufacturers can improve their products or recommend suitable products to their customers.

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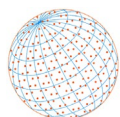
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SUPPLEMENTARY MATERIAL

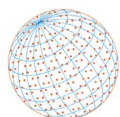
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